Inflammation and inflammatory mediators

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Overview

- **Inflammation** is a **normal response** of vascularized tissues to **infections** and **tissue damage** that **brings** cells and molecules of host defense from the **circulation** to the **sites where they are needed**, to **eliminate** the **offending agents**.
- It **serves** to **rid the host of** both the initial cause of cell injury (e.g., microbes, toxins) and the consequences of such injury (e.g., necrotic cells and tissues) .
- The **mediators (cells and molecules)** of defense include phagocytic leukocytes, antibodies, complement proteins, cytokines….etc.
- **Many immune cells and their precursors normally circulate** in the blood or reside in lymphoid and peripheral tissues in a resting or inactive state, so they **cannot damage normal tissues** but can be rapidly recruited to any site in the body when there are threats.

- Inflammatory mediators (i.e. molecules) are produced by resident cells, recruited cells, also produced from plasma proteins (eg. C3a, C3b,…etc.) that react to the microbes or to products of necrotic cells.

- Some of these **mediators** promote the efflux of plasma and the recruitment of circulating leukocytes to the site where the offending agent is located.

- **Some other mediators** (eg. Cytokines…etc.) also activate the recruited leukocytes enhancing their ability to destroy and remove the offending agent.

Understanding the role of mediators is important because most anti-inflammatory drugs target specific mediators….eg. anti- TNF-α.

- **Inflammation** is **terminated** when the **offending agent** is **eliminated**. The **reaction resolves because** mediators are broken down and dissipated, and leukocytes have short life spans in tissues. Likewise, anti-inflammatory mechanisms are activated, serving to control the response and prevent it from causing excessive damage to the host.

  - Without **inflammation**, **infections** would go **unchecked**, **wounds** would **never heal**, and **injured tissues** might remain **permanent festering sores**.
  - **Defective inflammation** is also responsible for serious illness **for example**, any compromise of marrow function will diminish the generation of mature leukocytes.

- Inflammation may be of two types:- Acute and chronic.
- Inflammation can be local or systemic.

What about sub-acute inflammation?

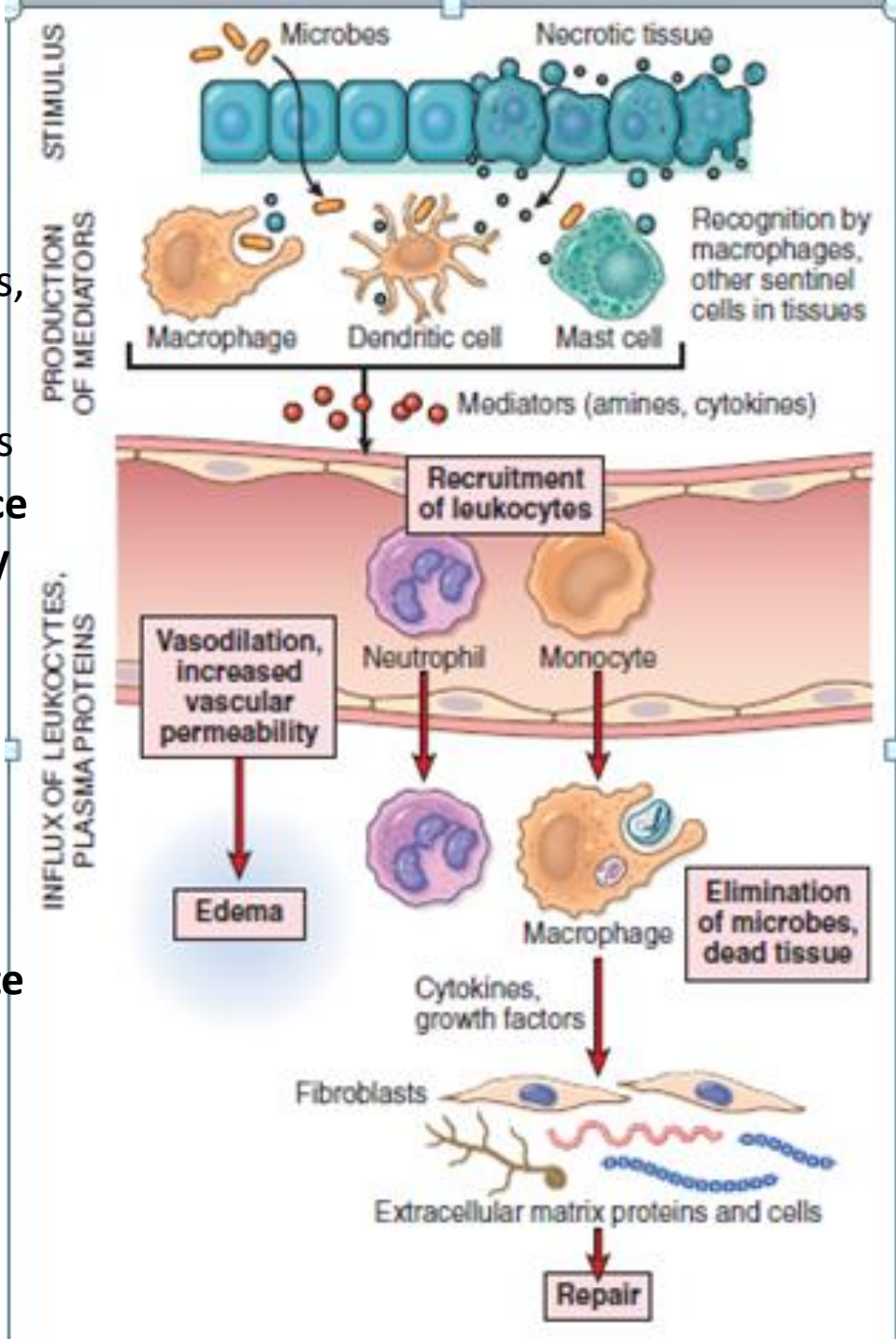
Causes and Triggers of inflammation

Inflammatory reactions may be triggered by a variety of stimuli:

- **Infections….** (bacterial, viral, fungal, parasitic) and microbial toxins.
- **Tissue necrosis** elicits inflammation regardless of the cause of cell death, which may include ischemia, trauma, physical and chemical injury (e.g., thermal injury, as in burns; irradiation; exposure to some environmental chemicals).
- **Foreign bodies….** (splinters, dirt)
- **Immune reactions…..** autoimmune diseases, allergies.

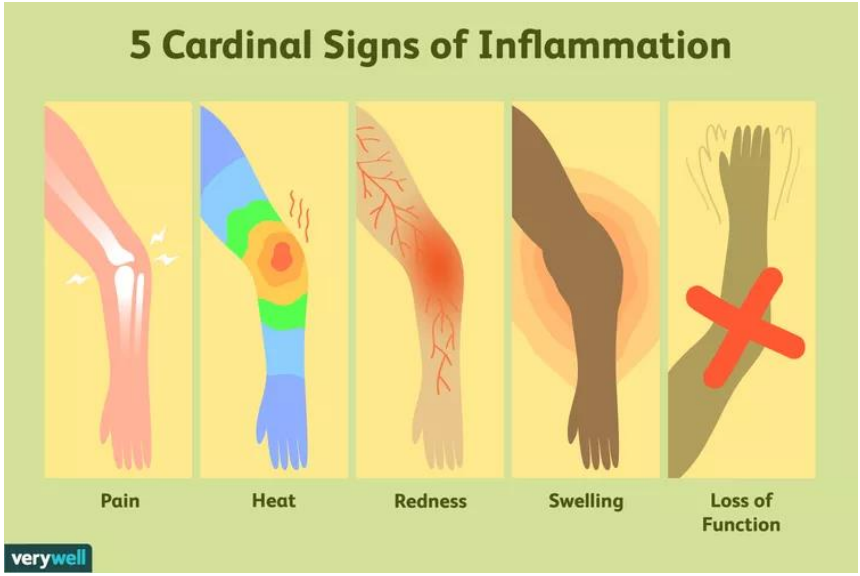
Inflammatory steps

- When **a microbe enters a tissue or the tissue is injured**, the presence of the infection or damage is sensed by resident cells, including macrophages, dendritic cells, mast cells, and other cell types.
- These cells **secrete** inflammatory molecules (**cytokines** and other mediators) that **induce** and **regulate** the subsequent **inflammatory response**.
- **Leukocytes and plasma proteins** are recruited from the **circulation** to **the site** where the **offending agent** is located.
- The **leukocytes and proteins** are activated and work together to **destroy and eliminate** the offending substance.
- The reaction is **controlled and terminated**.
- The **damaged tissue** is **repaired**.



The external manifestations of inflammation, often called its cardinal signs, are :-

- Heat (calor in Latin),
- Redness (rubor)
- Swelling (tumor)
- Pain (dolor)
- Loss of function (functio laesa).





✓ **Acute inflammation** is the initial, rapid response to infections and tissue damage. It typically develops within minutes or hours and is of **short duration**, lasting for several hours or a few days (eg. acute bacterial pneumonia).

✓ If the initial response **fails** to clear the stimulus, the reaction progresses to a **prolonged** type of inflammation

✓ **chronic inflammation**

✓ **Chronic inflammation** may **follow** acute inflammation or **arise de novo**. It is **of longer duration** and **is associated with** more tissue destruction, the presence of lymphocytes and macrophages, the proliferation of blood vessels, and fibrosis…

Examples….. rheumatoid arthritis, atherosclerosis, and lung fibrosis…etc.

Recognition of microbes and damaged cells

- The **first step** in inflammatory responses is the recognition of microbes and necrotic cells by **cellular receptors and circulating proteins.**

- **Cellular receptors for microbes**. Phagocytes, dendritic cells (in epithelia and all tissues whose function is to capture microbes), and many other cells **express receptors** that **detect** the presence of **infectious pathogens… (pattern recognition receptors (PRRs).**



Are proteins capable of recognizing molecules frequently found in pathogens (Pathogen-Associated Molecular Patterns—PAMPs like LPS), or molecules released by damaged cells (Damage-Associated Molecular Patterns—DAMPs like **uric acid** and **ATP).**

- The best defined of these **PRRs** belong to the family of **Toll-like receptors (TLRs)** Located in plasma membranes and endosomes, so they are **able** to **detect extracellular** and **ingested microbes**.

- Other example is **Nucleotide-binding oligomerization domain-like receptors (NLRs)** which are **cytoplasmic sensors** that **detect** intracellular bacterial components and cellular.

- **Recognition** of **microbes** by these **receptors** stimulates the production and expression of a number of secreted and membrane proteins like **cytokines** that induce inflammation, anti-viral cytokines (interferons), and cytokines and membrane proteins that promote lymphocyte activation or differentiation (eg. IL-12, IL-2).

- **Circulating proteins**: Several plasma proteins recognize microbes and function to destroy microbes and to stimulate inflammation at tissue sites of infection like **complement system**.

Acute inflammation

✓ Acute inflammation is a common result of innate immune responses, and local adaptive immune responses can also promote inflammation.

✓ **Acute inflammation has three major components:-**

❖ Dilation of small vessels, leading to an increase in blood flow and is induced by the action of several mediators, like histamine, nitric oxide, prostaglandins.

❖ Increased **permeability** of the microvasculature, enabling plasma proteins and leukocytes to leave the circulation. This is induced by histamine, kinins…etc

❖ **Emigration** of the leukocytes from the microcirculation, their accumulation in the focus of injury, and their activation to eliminate the offending agent.

***The most important **leukocytes** in typical inflammatory reactions are the ones capable of **phagocytosis**, namely, **neutrophils** and **macrophages** ingesting and destroy bacteria and other microbes, as well as necrotic tissue and foreign substances.

- Macrophages also produce growth factors that aid in repair like vascular endothelial growth factor (VEGF).
- Th17 cells enhance neutrophil and monocyte recruitment.

*** The journey of **leukocytes from** the vessel lumen **to** the tissue is a **multistep process** that is **mediated and controlled by** adhesion molecules and cytokines which **process can be divided into phases:-**

1. **Adhesion of leukocytes to endothelium** at the site of Inflammation. This is mediated by complementary **adhesion molecules (selectins and integrins**) on the two cell types **(leukocytes and endothelial cells)** whose expression is enhanced by cytokines that are secreted by cells in tissues in response to offending agents.

2. **Transmigration of the leukocytes** through the vessel wall, and movement of the cells toward the offending agent.

Selectins and Integrins

- **Selectins** mediate the initial **weak interactions** between **leukocytes and endothelium.**

- **E-selectin**,…on endothelial cells
- **P-selectin**, …. present on platelets and endothelium
- **L-selectin**….. Found on the surface of most leukocytes.

- **Integrins** mediated **firm adhesion** of leukocytes to endothelium.

- **TNF and IL-1,** activate endothelial cells to increase their expression of ligands for integrins. These ligands include intercellular adhesion molecule-1 (ICAM-1), and vascular cell adhesion molecule-1 (VCAM-1).

3. Transmigration leukocytes in the tissue

✓ After **exiting** the **circulation**, leukocytes move in the **tissues** toward the site of injury by a process called **chemotaxis**.

✓ Both exogenous and endogenous substances can act achemoattractants, including the following:-

- Bacterial products
- Cytokines, especially those of the chemokine family (eg. IL-8).
- Components of the complement system, particularly C5a

✓ These chemoattractants are **produced** by microbes and by host cells in response to infections and tissue damage and during immunologic reactions.

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Just to understand the process

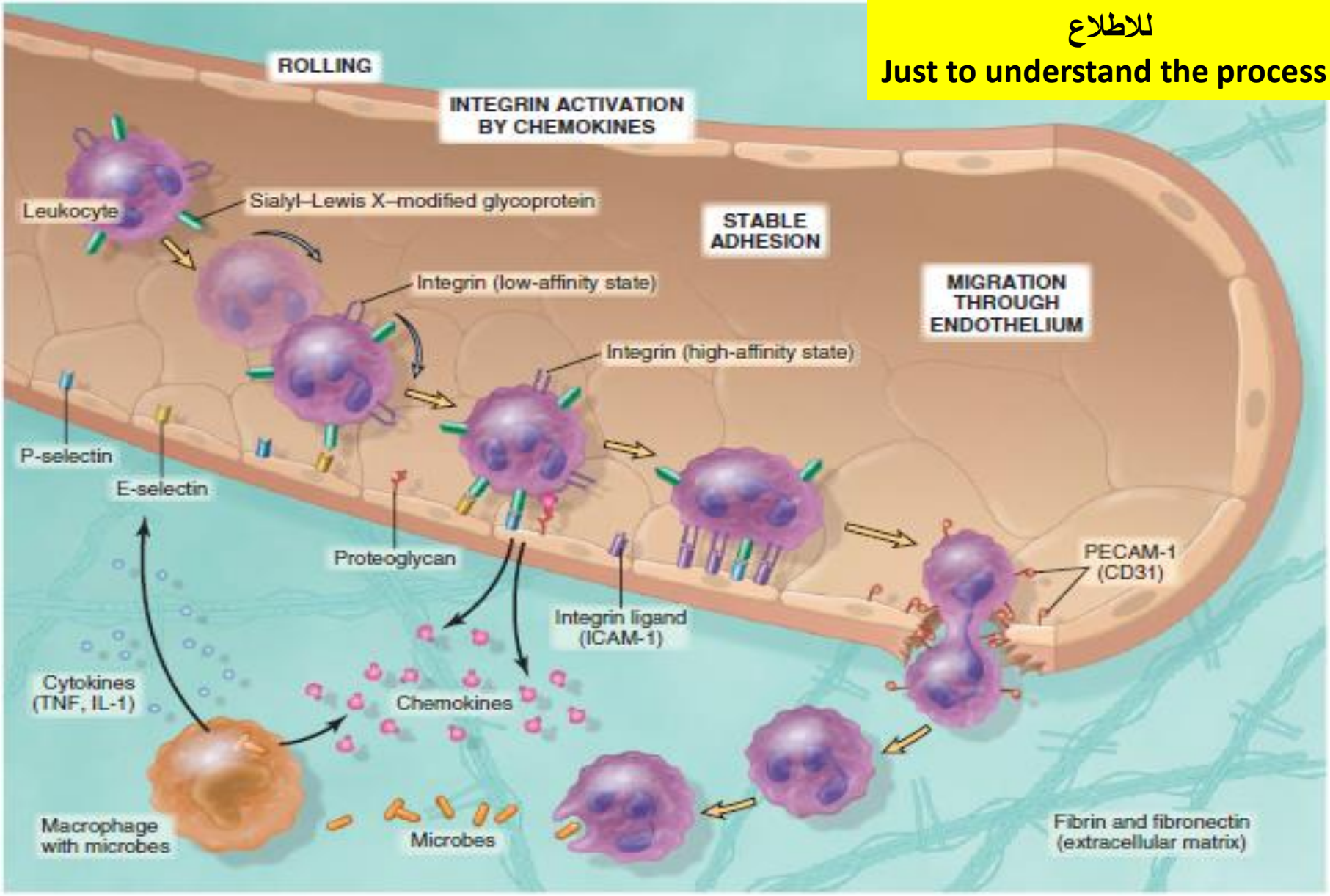


Fig. 3.4 The multistep process of leukocyte migration through blood vessels, shown here for neutrophils. The leukocytes first roll, then become activated and adhere to endothelium, then transmigrate across the endothelium, pierce the basement membrane, and move toward chemoattractants emanating from the source of injury. Different molecules play predominant roles at each step of this process: selectins in rolling; chemokines (usually displayed bound to proteoglycans) in activating the neutrophils to increase avidity of integrins; integrins in firm adhesion; and CD31 (PECAM-1) in transmigration. *ICAM-1*, Intercellular adhesion molecule-1; *PECAM-1 (CD31)*, platelet endothelial cell adhesion molecule-1; *TNF*, tumor necrosis factor.

- ✓ In most forms of **acute inflammation**, neutrophils predominate in the inflammatory infiltrate during the first **6 to 24 hrs** and are gradually **replaced** by macrophages over **24 to 48 hrs.**

- ✓ After entering tissues, neutrophils are **short-lived**; they undergo apoptosis and disappear within 24 to 48 hours.

- ✓ Macrophages not only **survive longer** but also may proliferate in the tissues, and thus they become the dominant population in **prolonged** inflammatory reactions.

Phagocytosis and cytokine production

There are, however, exceptions to this pattern of cellular infiltration

- In certain infections—for example, those produced by *Pseudomonas* bacteria—the cellular infiltrate is dominated by neutrophils for several days.

- In viral infections, lymphocytes may be the first cells to arrive.

- Some hypersensitivity reactions are dominated by activated lymphocytes, macrophages, and plasma cells (reflecting the immune response).

- In allergic reactions, eosinophils may be a prominent cell type.

Leukocyte-Mediated Tissue Injury

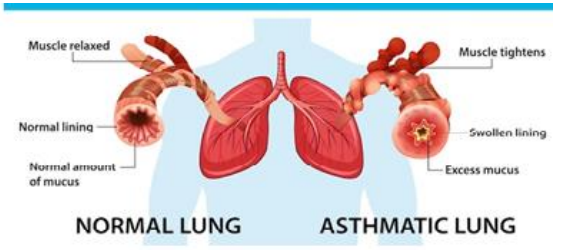
- In some infections that are difficult to eradicate, such as **tuberculosis** and **hepatitis**, the **prolonged host response** contributes more to the pathology than does the microbe itself.



- When the **inflammatory response** is inappropriately directed against **host tissues**, as in certain **autoimmune diseases**.


rheumatoid arthritis

- When the host hyper-reacts against usually harmless environmental substances, as in **allergic diseases**, including **asthma**, and **some drug reactions**.



Termination of the Acute Inflammatory Response and resolution

- **Anti-inflammatory mediators** terminate the acute inflammatory reaction when it is no longer needed for example:- **IL-10….** anti-inflammatory cytokine. Transforming growth factor-β (**TGF-β**)….. participates in tissue repair.
- Decreased inflammatory mediator production
- Cessation of neutrophil recruitment
- Neutrophil apoptosis
- Macrophage efferocytosis
- Clearance of debris
- Tissue repair

Chronic inflammation

- Is a **response of prolonged duration** (weeks or months) in which inflammation, tissue injury, and attempts at repair coexist, in varying combinations.
- Is mediated predominately by macrophages lymphocytes, plasma cells, and other leukocytes.
- Cytokines produced by macrophages and lymphocytes (notably T lymphocytes) are mostly involved.

Causes of Chronic Inflammation

- **Persistent infections** by microorganisms that are difficult to eradicate, such as mycobacteria.
- **Unresolved acute inflammation** evolves into chronic one, such as when an acute bacterial infection of the lung progresses to a chronic lung abscess.

- **In allergic diseases**, as in bronchial asthma. Such diseases may show morphologic patterns of mixed acute and chronic inflammation.

- **Prolonged exposure to potentially toxic agents**, either exogenous or endogenous.

Example
Atherosclerosis thought to be induced, at least in part, by excessive production and tissue deposition of endogenous cholesterol and other lipids.

Example
agent is particulate silica, results in an inflammatory lung disease called silicosis

- **In autoimmune diseases** like rheumatoid arthritis in which immune response induce against self and non self Ags.

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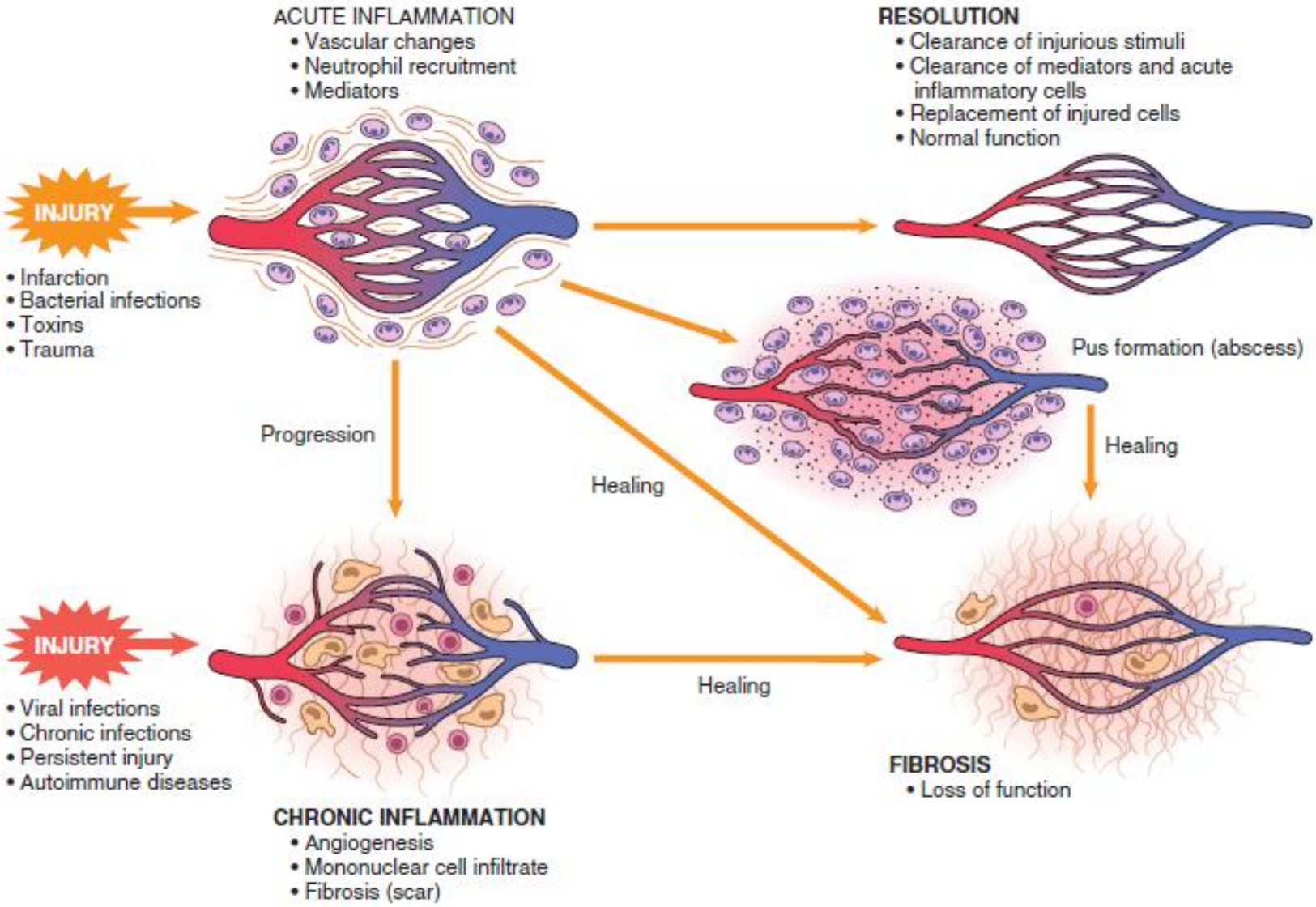


Fig. 3.16 Outcomes of acute inflammation: resolution, healing by fibrosis, or chronic inflammation. The components of the various reactions and their functional outcomes are listed.

The acute-phase response consists of several clinical and pathologic changes:

- ✓ **Fever:-** Substances that induce fever are called pyrogens like TNF .

- ✓ **Acute-phase proteins:** Are plasma proteins, their plasma concentrations may increase several hundred-fold as part of the response to inflammatory stimuli. **Examples**:-

1. C-reactive protein (CRP)
2. Fibrinogen
3. Serum amyloid A (SAA) protein.

- ✓ **Leukocytosis** is a common feature of inflammatory reactions, especially those induced by bacterial infections.

- ✓ **Sepsis and septic shock**: In some severe infections, Fall in blood pressure, disseminated intravascular coagulation, metabolic abnormalities, induced by high levels of TNF and other cytokines

Repair by scar formation

- **Repair** occurs by deposition of connective tissue and scar formation if the injured tissue is not capable of regeneration.

- **Macrophages** are critical for orchestrating the repair process, by eliminating offending agents and producing cytokines and growth factors that stimulate the proliferation of the cell types involved in repair.

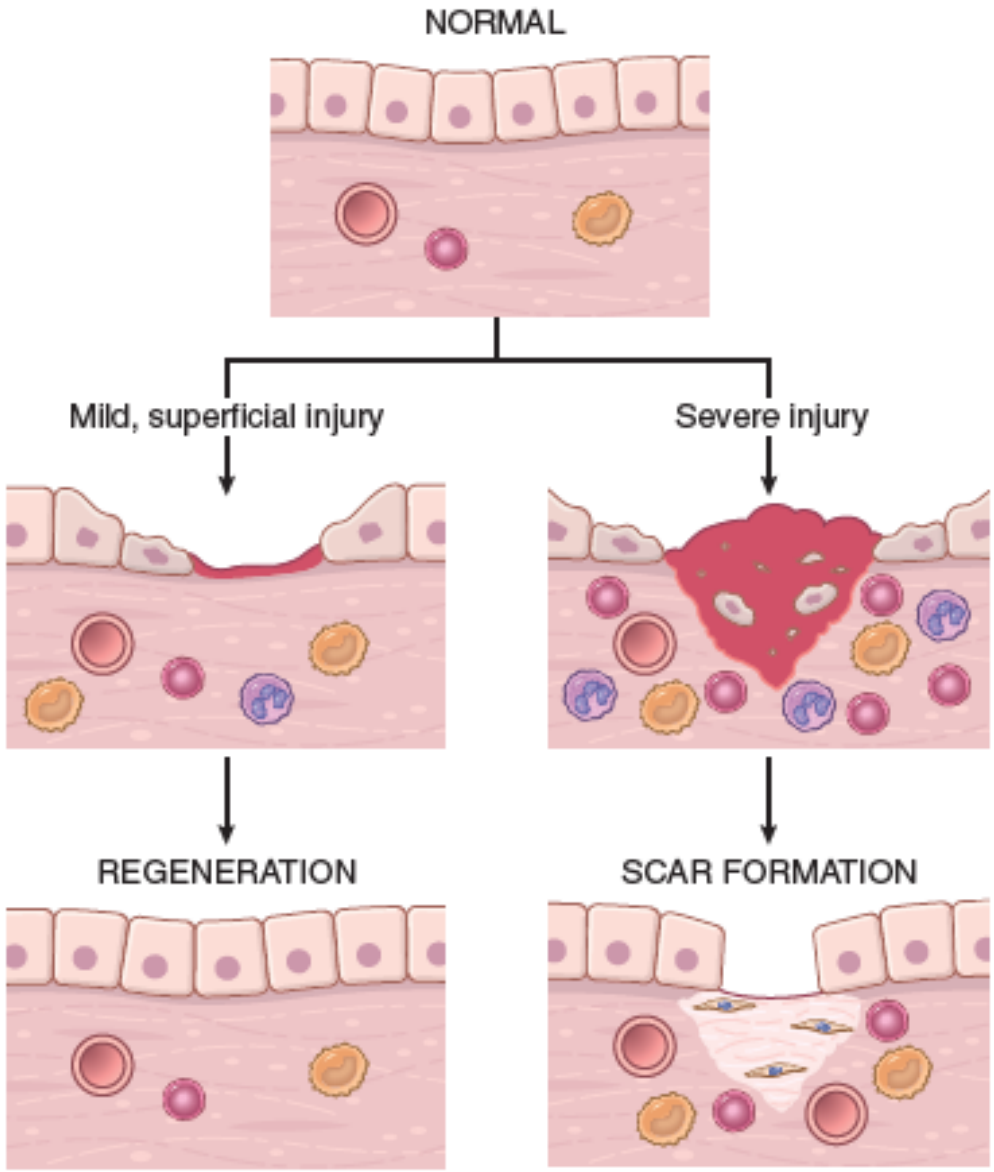


Fig. 3.23 Mechanisms of tissue repair: regeneration and scar formation. Following mild injury, which damages the epithelium but not the underlying tissue, resolution occurs by regeneration, but after more severe injury with damage to the connective tissue, repair is by scar formation.

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Just to understand the process

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